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# Low Phosphorus Electroless Nickel Coating Technology

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*SUMMARY — The development and advantages of low (1-5% phosporus) phosphorus electroless nickel coating processes is reviewed. Many applications are envisaged for these deposits.*

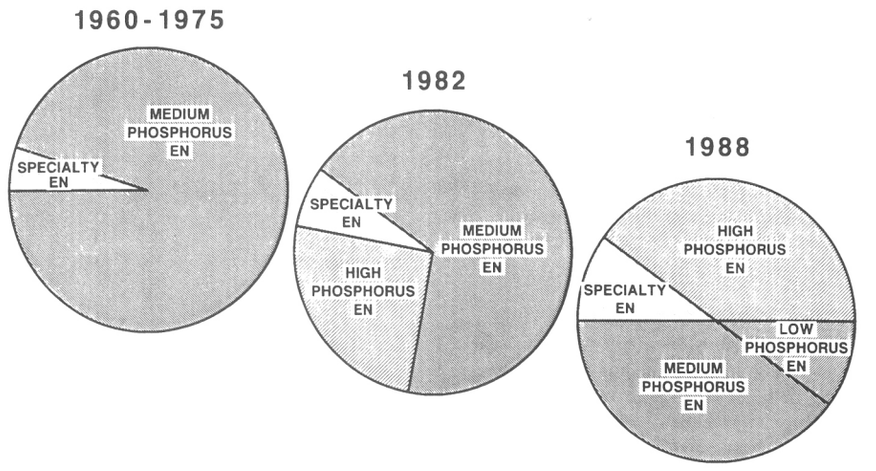
The evolution of electroless nickel technology in Europe and the U.K. over the last 25-30 years has been somewhat different than in the United States. A fundamental reason for these differences has been the markets and applications which comprise these geographies.

Historically, electroless nickel in the U.S. had its origins with the commercialisation of the Kanigen process by General American Transportation (GATX). This process provided a deposit of about 8-10% phosphorus and was particularly suited for very large, engineering plating operations. In fact, the first of commercial applications were for plating storage vessels for nuclear plants and for lining new tank cars produced by GATX. The Kanigen process then found its way into a number of other markets including aerospace, food processing, chemical, steel, etc, with varying degrees of success. In general, however, the initial market exposure for electroless nickel was broad based and resulted in a mixture of failure and success.

Proprietary electroless nickel process, suitable for use in a job shop plating operation, entered the U.S. market in the early 1960's. In general, these products were medium phosphorus deposits that provided only the basic attributes of the electroless nickel process and its generic deposit characeristics. During the next 15 years or so, improvements focused mainly on bath performance and not as much on the area of deposit properties and engineering related performance issues as required for specific markets. As a result, limitations existed in corrosion protection, enhanced wear and abrasion resistance, fatigue life, etc., in the electroless nickel coatings needed to satisfy more sophisticated applications. In the late 1970's and into the 1980's, substantial emphasis was placed on the increased use of high phosphorus electroless nickel coatings. This was due to the improved corrosion resistance, compressive stress and improved fatigue life, non-magnetic characteristics and the increased demand for these performance characteristics in important emerging markets. As indicated in Figures 1 and 2, the shift in the relationship between markets from the 1960's through the 1980's and the associated market share for conventional versus high phosphorus electroless nickel coatings was significant. Since the early 1980's, emerging engineering electroless nickel process development and the use of specialised deposits to satisfy individual market requirements has significantly expanded market penetration within the U.S. In addition, a higher rate of application success has accompanied this growth as users have developed more understanding of electroless nickels.

In Europe and in the U.K., however, the early use of electroless nickel appeared

*Figure 1. U.S. Market and process trends in electroless nickel plating.*




B. Jackson, R. Macary and G. Shawhan, Trans. Inst. Metal Finish., 1990, **68**(3), 75

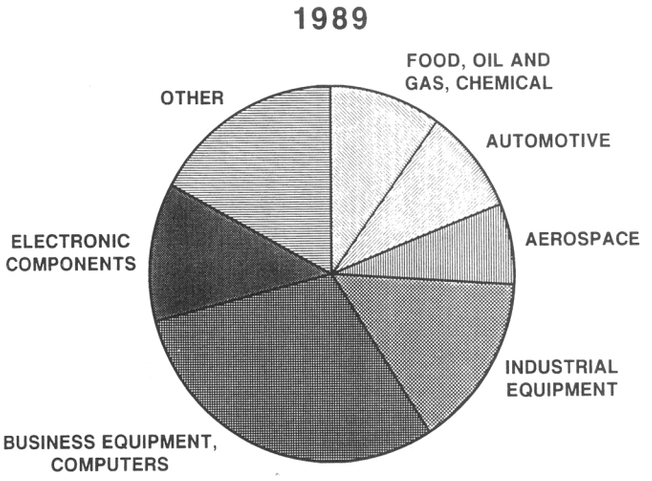



*Figure 2. Distribution of key markets for electroless nickel.*

more directed at demanding engineering applications, particularly those that required excellent wear and abrasion resistance. In Germany, for example, the principle process for many years was nickel boron rather than nickel phosphorus. The nickel boron process introduced by Dupont which contained thallium was primarily targeted towards high wear, abrasion, and erosion applications such as those found in the aerospace, automotive, and textile industries. Interestingly, electroless nickel boron never made an impact on the U.S. market except in very specialised market segments such as electronics, principally for its solderability, higher melting temperature and as a gold alternative. These were low boron content (<1%) products and not the 3-5% boron containing engineering coatings which possess significant benefits in hardness, wear, and abrasion resistance over conventional nickel phosphorus alloys.

The marketplace in Europe and the U.K. was comprised of key industries (such as Textile, Molds, Mining, Automotive, and Industrial Machinery) that were perhaps more sophisticated and demanding in their technical requirements than those industries in the U.S. which were considering electroless nickel. In addition, these industries were more focused on the use of electroless nickel as an improved process for replacing such finishes as hard chromium and certain other finishes for wear and abrasion resistance. During the emerging years of electroless nickel, a substantial amount was learned about the successful use of these coatings for these purposes in the U.K. and Europe rather than in the U.S. The importance of electroless nickel for use as wear resistant coatings has continued to be an important focal point for new applications. Furthermore, perhaps due to the nature of the U.K. and European markets, leadership has continued to be shown in the introduction and commercial use of processes such as electroless nickel with PTFE to solve wear and friction problems.

The recent development of low phosphorus electroless nickel has produced an exciting new coating technology which has significant deposit and process advantages over conventional medium and high phosphorus electroless nickel coatings. Furthermore, these coatings provide an important alternative to nickel-boron due to their similar engineering properties, reduced waste treatment problems, and generally easier to operate plating chemistry. Clearly the nature of the key markets for electroless nickel in the U.K. and Europe makes the availability of low phosphorus technology an important new technology.

Low phosphorus electroless nickel technology is characterised by unique engineering properties not found in other types of electroless nickel phosphorus deposits. As an arbitrary point of definition, these products can contain 1-5% by weight phosphorus and by their formulation, exhibit higher as-plated hardness than other nickel phosphorus deposits. It is, in fact, possible to produce lower phosphorus deposits from processes without such significant improvements in deposit performance, so some caution should be exercised with regard to overgeneralisation of the term "low phosphorus". Also, it should be noted that other properties such as corrosion protection, stress and fatigue can be affected by formulation differences while the basic hardness and wear properties are retained. In comparison, a medium phosphorus deposit is generally categorised as having 5-8% phosphorus and high phosphorus electroless nickel deposits as containing 9-12% phosphorus by weight. Again, the other principle difference between these deposits and low phosphorus is in the area of hardness and wear while other differences in deposit properties may vary for a variety of reasons related to bath chemistry.

## WEAR RELATED PROPERTIES

Let us first examine the wear related properties of electroless nickel and how low phosphorus deposits compare, as suggested in Table 1. Clearly this type of deposit offers significant improvements over other nickel phosphorus alloys and gives a more comparable performance to boron nickel alloys.

### Hardness

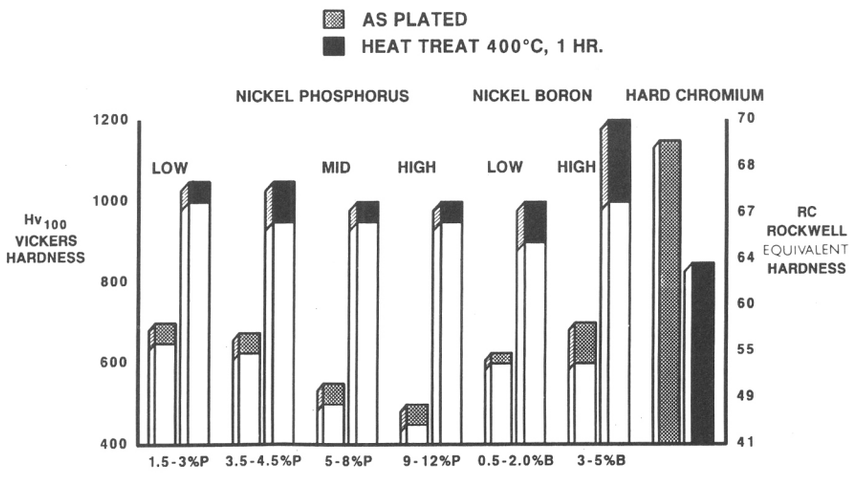
The Vickers hardness of electroless nickel deposits as measured by the ASTM method or DIN method requires a deposit thickness of 75μm minimum. This is, of course, necessary to avoid major substrate influences in the measurement. As indicated in Figure 3, the hardness values of two low phosphorus deposits of 1-3%P and 3.5-4.5%P by weight are superior to alternate nickel phosphorus alloys and provide a definite alternative to a 3-5% by weight nickel boron coatings. The fact that the as-plated hardness of the 1-3% low phosphorus nickel deposit is in the range of 700

**Table 1. Electroless Nickel Technology Engineering Guidelines for Wear Related Properties**

| | *MP* | *HP* | *LP* | *NI-PTFE* |
|---|---|---|---|---|
| Fretting wear | G | G | S | P |
| Wear/galling | | | | |
| Low load | G | G | S | F/G/S |
| High load | F/G | F/G | S | P |
| Cyclical load | G | G | S | P |
| Erosive wear | G | G | S | P |
| Fatigue wear | P/F | P/F | S | P |
| Sliding wear | | | | |
| Dry | F/G | F/G | G/S | S |
| Lubricated | G | G | S | S |
| Friction | G | G | G | S |
| Abrasive wear | F/G | F/G | S | P |

S = Superior G = Good F = Fair P = Poor

*Figure 3. Typical Vickers hardness of selected electroless nickel deposits "100g load"*



Vickers means very hard deposits can be applied to heat sensitive metals like aluminium without having to be heat treated. Furthermore, the cost of heat treatment can be eliminated in many cases since the as-plated hardness is so high. Heat treated low phosphorus deposits approach hard chromium hardness values in the as-plated condition and exceed their hardness when hard chrome is heat treated. Thus LP electroless nickel coating offers a very good alternate to hard chromium due to its improved hardness.

**Abrasion**

The Taber Abraser Wear Tester is a well recognised procedure for evaluating the abrasion resistance of coatings under a constant load. Abrasive wear can be characterised more commonly as non-lubricated adhesive wear. In this instance, sharp metal or other abrasive particles cut into a mating metal surface. This method is defined by ASTM and DIN. The common result reported from this test is the TWI (Taber Wear Index), which is the milligrams of weight loss per 1000 cycles. The initial 1000 cycles is eliminated from the final calculation due to break-in effects. A CS-10 wheel and an external load of 1000 g are typically specified for electroless nickel deposits. As indicated in Figure 4, the low phosphorus deposit has far superior abrasion resistance to alternate electroless nickel phosphorus deposits and favourably compares to hard chromium or high boron nickel coatings. This is very significant in that the advantages of the electroless process can be accessed by design engineers faced with constraints from finishes such as hard chromium or hard anodizing. Furthermore, significant improvement in the abrasion resistance of electroless nickel can be realised using these new coatings.

**Adhesive Wear and Galling**

Adhesive wear, in a lubricated environment, is normally associated with mating metal surfaces that develop a boundary condition. In this instance, asperities on one moving surface come in contact with the other surface, breaking through the hydrodynamic lubricant film which exists during periods of inadequate lubrication. Heat is immediately generated under boundary conditions and if the lubricant film is not restored, galling eventually results. The mechanism in adhesive wear is the generation of heat and formation of adhesive bonds between the two moving surfaces. These bonds fracture during continued motion causing metal particles to be pulled from one surface and then cause wear or eventually galling. The potential for adhesive wear is influenced by several factors including load, lubricant load, lubricant type, coating type and hardness differential between the two mating surfaces. In the case of load, there may be low loads, high loads, point loads and/or cyclical loads. In general, the higher the load and/or the wider and more frequent the shift in the load during operation, the greater the chance that the hydrodynamic film will be breached and adhesive wear will occur. Under such conditions, the lubricant, and the hardness of the mating surfaces play important roles in determining the ability of the interacting metal surfaces to "slip-off" each other. In the case of the metal surfaces, including plated metal surfaces, the hardness differential is very important in optimising the wear pair and avoiding the potential for galling. For optimising wear pairs, low phosphorus electroless nickel deposits are expected to perform their best under conditions of increased load or point load due to the improved load support they provide. Also, LP deposits should have less tendency to work harden locally under such loads due to their lower phosphorus content and higher initial hardness. They will, however, present a potential problem if they contribute to a significantly increased hardness differential compared to the mated part. High phosphorus electroless nickel, which is softer, can give better performance under such conditions. In contrast, more highly alloyed high phosphorus deposits will have a greater potential to work harden under boundary conditions creating a localised hard surface with a soft underlayer. If this occurs under conditions of higher load or cyclical load, galling can again result as a consequence of the hard surface pushing into the soft underlayer.

A general attribute of electroless nickel is its "good lubricity" or ability to dissipate some heat as contact is made between two mating surfaces. Anti-wear lubricants are designed to react at the point of contact when boundary lubrication conditions are established, causing the two surfaces to slip off each other and back into a hydrodynamic condition. Electroless nickel plated metals may assist in accomplishing this result by helping to support the lubricant film better. This may be accomplished by allowing the lubricant to wet the surface better. Certainly, this is true when electroless nickel is plated onto metals commonly prone to galling and adhesive wear failures such as stainless steel, aluminium or titanium. Furthermore, electroless nickel can aid in reducing adhesive wear due to its uniform thickness which avoids having non-uniform areas on the surface that create localised loading and the resultant potential for boundary lubrication. Adhesive wear tests of several types have been developed as a means of evaluating these performance properties. The Falex and Alpha LFW-1 wear tests are important wear tests for this purpose. As shown in Figure 5, the resistance of electroless nickel deposits in general to adhesive wear is very good in the Falex test and the performance of the low phosphorus electroless nickel is superior to the other electroless nickel phosphorus deposits tested.

*Figure 4. Taber abraser wear testing of several electroless nickel deposits; CS-10 wheel, Taber wear index, mg loss/1000 cycles.*

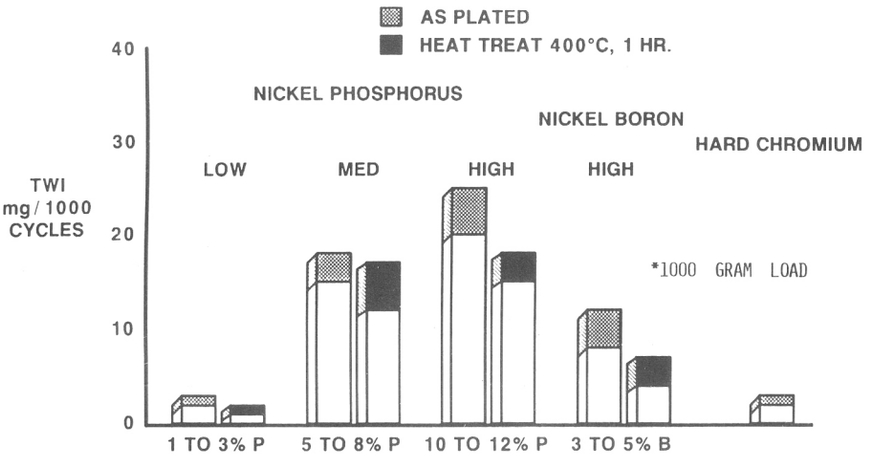
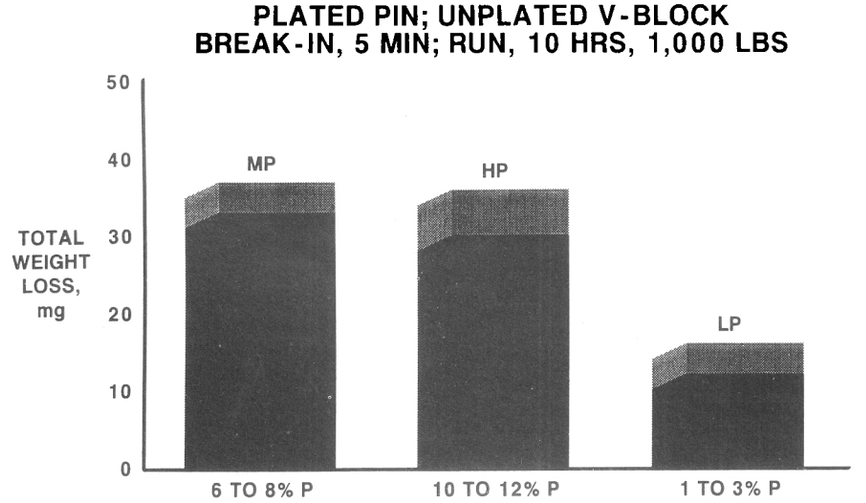



*Figure 5. ASTM D2625 Falex wear performance of several electroless nickel-phosphorus deposits.*




**Fretting Wear**

An important application for electroless nickel is in solving problems with fretting wear. Fretting wear is developed as a consequence of the oscillatory motion between two mating surfaces. A typical example of this type of wear is an external fretting of an

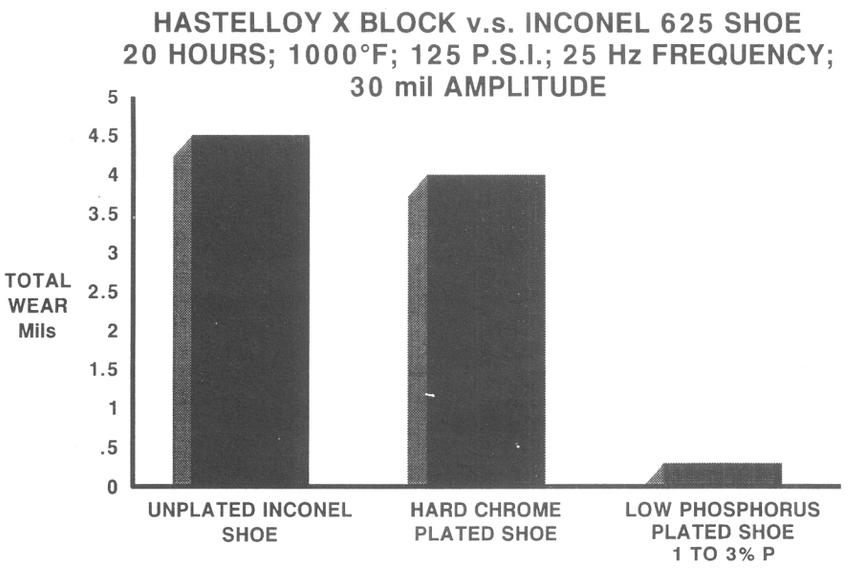



*Figure 6. Fretting wear test*

aircraft fuel line fitting which is subject to high vibration. Normally no lubrication is present under such conditions. The constant vibration during operation generates heat, breaks loose particles and causes a localised abrasive action or polishing action to occur between two surfaces. Higher hardness deposits capable of dissipating heat at the points of contact are desirable in dealing with this kind of wear.

Fretting wear tests have been conducted on several electroless nickel deposits and compared to hard chromium. As indicated in Figure 6, the low phosphorus electroless nickel deposit is superior in its fretting wear properties. This advantage is attributed to the greater ability of LP technology to dissipate thermal energy during metal to metal contact. Furthermore, its compressive stress and low alloy content make it less prone to chipping, spalling or work hardening.

**Fatigue Wear**

Fatigue wear is normally associated with bearing applications where load is applied to a rotating or rolling cylindrical surface under lubricated conditions. During continued operation with load shifts, the cyclical applied stress can develop fatigue cracks which propagate until failure. The fatigue failure can be fast once the crack propagation mechanism starts. Electroless nickel, in general, can be an effective coating to help reduce this problem. In particular, a compressively stressed deposit which can support cyclical loads, is desirable and suggests that LP electroless nickel may be particularly good for coating mating surfaces of bearings.

**Erosive Wear**

Erosion is directly related to impinging particles which impact a surface at a given velocity. Impingement angle, temperature conditions, and particles of differing shapes and sizes, all influence the erosion rate. The mechanism of failure varies since at low impingement angles, ie 20°, the metal coating is removed by a cutting action. Thus compressively stressed coatings that "give-a-little" under impact are more desirable. At a high angle of attack, ie., 90°, hardness is an important criteria since chipping and spalling are a key mechanism of failure along with crack propagation toward the substrate. These are not the only issues, however. A very important aspect to erosion resistance is toughness or the ability to dispel thermal energy generated at the point of contact.

Electroless nickel coatings, in general, have been shown to provide good performance in erosion resistance, particularly those which have a compressive stress. Importantly, LP electroless nickel deposits have shown noticeably better erosion resistance. This, in fact, is attributed to their higher melting temperature and greater ability to dissipate the thermal energy developed during impact. Results of erosion tests on electroless nickel run at a 20° impingement angle are given in Figure 7. In this test a 325 mesh alumina grit was used at a pressure of 60 psi.

**Stress**

Electroless nickel deposits vary from tensile to compressive stress depending upon the bath chemistry including the age of the process solution. Typically stabilisers, other additives, by-products from continued plating (i.e., orthophosphite) and perhaps phosphorus content, influence the stress of the deposit. High phosphorus electroless nickel deposits which utilise additives that avoid tensile stresses, have been the principle compressively stressed electroless

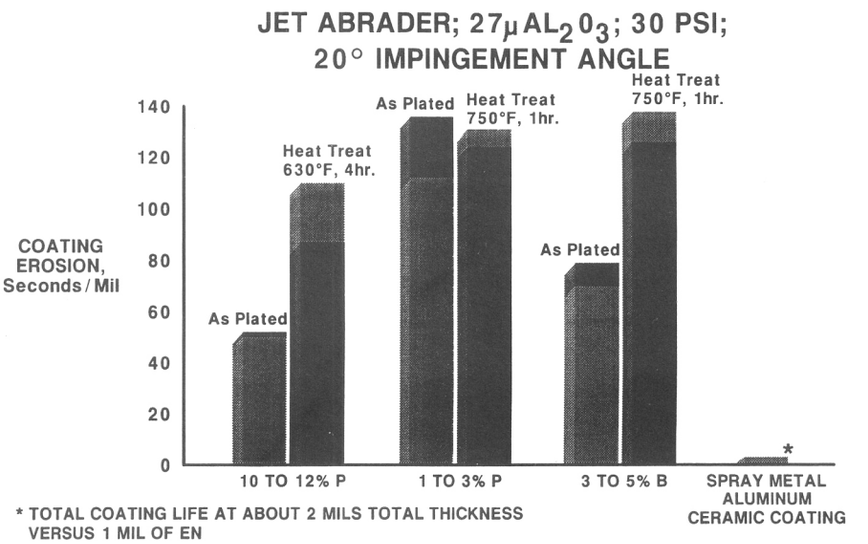
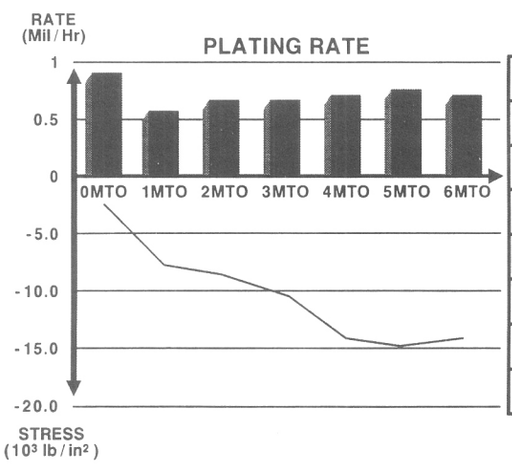



*Figure 7. Erosion testing of several electroless nickel coatings vs. competitive processes.*

*Figure 8. Compressive stress of 1 to 3% P low phosphorus EN versus bath life.*




| MTO | pH | T°F. | RATE (Mil / Hr) | S 10³ (lb / in²) |
|---|---|---|---|---|
| 0 | 6.4 | 180 | 0.85 | -2.73 |
| 1 | 6.4 | 180 | 0.68 | -7.80 |
| 2 | 6.4 | 180 | 0.65 | -8.50 |
| 3 | 6.4 | 180 | 0.62 | -11.75 |
| 4 | 6.4 | 180 | 0.67 | -14.30 |
| 5 | 6.4 | 180 | 0.68 | -14.45 |
| 6 | 6.4 | 180 | 0.65 | -14.20 |

nickel deposits available on the market. Depending upon the process, these coatings can stay on the compressive side for 3-6 or more bath turnovers (replenishment of original metal content). Low phosphorus electroless nickel coatings have shown a unique characteristic of remaining compressive in stress even beyond the six metal turnover point, as shown in Figure 8. It is important to note, however, this characteristic is based on overall bath chemistry and not just phosphorus content. This is another important advantage for engineers who are not only concerned about improved wear performance, but also avoiding stressed deposits that are prone to chipping, etc.

## ELECTRICAL PROPERTIES
## OTHER PHYSICAL PROPERTIES

Low phosphorus electroless nickel deposits possess several other important attributes that increase their usefulness for new market application. The electrical resistivity of low phosphorus 1-3%P electroless nickel has a typical range of 20-30 microohm-cm as-plated. Heat treated, the resistivity has been shown to fall to <15 microohm-cm. In both cases, these values are better than conventional electroless nickel and make low phosphorus deposit more suitable for contacts, mating surfaces of connectors, etc.

The higher melting temperature of the 1-3%P deposit has shown advantages in welding applications for metal package lids as an example, although the measured melting temperature has varied depending upon the test method. In general, values above 1090°C have been measured in certain cases. This advantage has demonstrated advantages in high temperature glass molds, for example, where conventional electroless nickel is not effective.

### Metallurgical Properties

The 1-3% deposit has been characterised metallurgically as a well defined lamellar structure using metallography. This structure after heat treatment at 400°C retains its lamellar structure. Under this condition, a limited precipitate of nickel phosphide has been identified using a wave length dispersive X-ray technique. At 600°C, a random well defined precipitate is evident.

Electron Spectroscopy for Chemical Analysis (ESCA) and Auger Electron Spectroscopy have also been used to characterise low phosphorus 1-3%P deposits. In the as-plated condition, the LP deposit surface showed a high percentage of nickel metal with little $Ni(OH)_2$. In contrast, a high phosphorus deposit has a very high percentage of $Ni(OH)_2$ and much less nickel metal. This layer is < 100Å thick. Perhaps most interesting is the fact that phosphorus was depleted in the $Ni(OH)_2$ region. Thus the low phosphorus deposit is higher in phosphorus at the surface while the high phosphorus deposit is depleted of much of its phosphorus at the surface.

Upon heat treatment at 400°C for one hour, the form of nickel on the deposit surface was NiO with no nickel metal being left at the surface. The measured depth of this oxide layer was 150-200Å.

## CORROSION PROPERTIES

One of the most intriguing revelations with regard to low phosphorus electroless nickel is its superior corrosion resistance on steel to high phosphorus electroless nickel in alkaline and certain other chemical environments. In contrast, acidic environments require pore free high phosphorus deposits to protect the substrate from the effects of galvanic corrosion.

### Porosity and Salt Spray Resistance

At the outset, it is essential that the relationship between deposit porosity and results from accelerated tests such as salt spray or CASS be reviewed.

The ASTM B-117 (DIN 50021) 5% neutral salt spray method for evaluating the corrosion performance of coatings has been used extensively and is a requirement in many commercial and military specifications. This test method is performed on plated flat panels with masked-off edges. The panels are placed in a chamber at a specific angle (15°-30°) and the effects of the salt environment evaluated on the top surface. Failure is normally defined by the hours to produce the first corrosion spot. The only critical surface in this case is the top of the panel, upon which the salt fog falls.

Salt spray studies have been conducted on 1010 carbon steel with deposits from several electroless nickel processes of differing bath chemistries and deposit compositions. Results of salt spray test (Table 2) on these deposits show several important relationships between different deposit types. First the performance of a HP deposit at a thickness of 12.5 microns (0.5 mils) is far superior to the MP and LP deposits 1 and 2 in the as-plated condition at thicknesses of up to 50 microns (2 mils). Furthermore, an increase in deposit thickness above a certain threshold level does not appear to compensate for inherent differences in the protective properties of alternative electroless nickel deposits. As suggested by the data in Table 3, a threshold thickness appears to exist (at least on relatively smooth surfaces) for a particular deposit in its ability to eliminate porosity to the basis metal, in the range of 18 to 25 microns (0.7 to 1.0 mils). It is important to remember that in this instance, the surface is relatively defect free and no edge areas are exposed. Thus, in this test, the HP deposit is able to achieve about 250 hours without failure compared to 24 hours for the other deposits. This is attributed to differences in the coverage ability of the deposit, which is influenced by phosphorus content, plating rate, bath chemistry, and possible differences in bath operation, such as the method of replenishment additions. Low phosphorus deposits vary in their ability to perform well in salt spray principally due to differences in bath chemistry.

Heat treatment of the HP deposit at 400°C (753°F) for 1 hour to achieve additional coating hardness, however, causes metallurgical changes (microcracking) in the deposit that result in a

**Table 2. ASTM B-117 (DIN 50021) Salt Spray Performance of Several Electroless Nickel Deposits on 1010 Carbon Steel Varying Thickness**

| | | | *Hours to First Corrosion Spot* | | | |
|---|---|---|---|---|---|---|
| | | | *Deposit A** | *Deposit B*** | *MP Deposit C**** | *HP Deposit D***** |
| | 12.5 | ... | 24 | >96 | 24 | 250 |
| Deposit | 25 | ... | 24 | >250 | 96 | 1000+ |
| Thickness, | | ... | −NR | −NR | 96 | 1000+ |
| Microns | | ... | −NR | −NR | 96 | 1000+ |

*Process A produces a 1-3 weight % LP phosphorus deposit, 12.5-20 micron/hr plating rate.
**Process B produces a 3.5-4.5 weight % LP phosphorus deposit, 20-25 micron/hr plating rate.
***Process C produces a 7-8 weight % MP phosphorus deposit, 17.5-20 micron/hr plating rate.
****Process D produces a 10-12 weight % HP phosphorous deposit, 7.5-12.5 micron/hr plating rate.

**Table 3. Materials Tested in Weight Loss Corrosion Tests**

| *Chemical* | *Typical Applications* |
|---|---|
| Thionyl Chloride | Chemical Process Intermediate |
| Orthochlorobenzyl Chloride | Agricultural Products, Pharmaceutical, Plastics |
| Phosphoric Acid | Food Processing, Pharmaceutical, Gelatins, Glues, Adhesives, Metal Processing, Fabrics, Plastics, Chemical Polishing |
| Benzotrichloride | Chemical Intermediate, Preparation of Dye Stuffs |
| Benzoyl Chloride | Perfumes, Chemical Intermediate, Resin Manufacture, Pharmaceuticals, Pesticide Intermediate |
| Phosphorus Oxychloride | Organic Phosphates, Plasticizers, Gasoline Additives, Preparation of Dye Stuffs, Medicines, Chemical Intermediate |
| Sodium Hydroxide | Food Processing, Petroleum Refining, Pharmaceuticals, Cosmetics, Bleaching, Dying, Printing, Textile Manufacture, Petroleum Refining |

noticeable reduction in its salt spray performance. These changes do not necessarily occur in the low phosphorus deposit as noted previously under the same conditions. As a general observation, the salt spray test method has shown its main value as a method of comparison for various coatings to help assess their relative porosity and little else.

ASTM B-117 (DIN 50021) has also been utilised frequently to evaluate the relative corrosion performance of electroless nickel on actual production parts. In this case, however, the correlation to actual production service has been poor and even resulted in incorrect conclusions. One important reason for this lack of correlation is the effect of the surface condition of production parts and it's contribution to porosity in the final electroless nickel deposit. These pores in the electroless nickel plating may or may not be significant to the anticipated service performance of the part. This is because the 5% neutral salt spray test does not take into account the dynamics of the actual corrosive environment which is an essential element in assessing the suitability of electroless nickel for a particular application. In general, when salt spray or accelerated test methods such as this are used, their utility and reliability is for drawing conclusions with respect to porosity and not corrosion resistance.

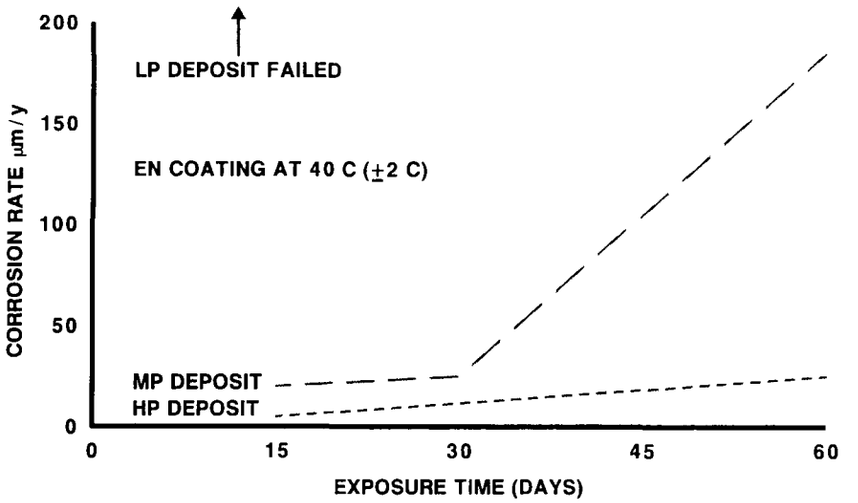



*Figure 9. Corrosion performance of 75% phosphoric acid 60 day weight loss test.*

**Corrosion Resistance**

Electroless nickel is gaining acceptance as a corrosion resistant coating in many process industries as a cost effective alternative to high alloy steels and various coatings or linings already in service. Food, meat, fish and poultry processing, oil and gas, chemical manufacturing, and the aerospace and aircraft industries are among the more important examples. These industries present a broad range of corrosion control problems. Applications in these process industries include environments that range from highly alkaline to highly acidic. In such environments, a particular electroless nickel deposit must be tested for determination of its corrosion resistance and/or corrosion protection capabilities. The following data show comparative performance results for low phosphorus, medium phosphorus and high phosphorus electroless nickel deposits in selected corrosion environments.

Process acids, such as hydrochloric acid, sulphuric acid, and phosphoric acid are encountered in many process industry applications. Food products, well completion, mining, chemical manufacture, textiles, and plastics are common examples. Weight loss corrosion testing performed over a 60-day period in phosphoric acid (75%) is presented in Figure 9 for the three different electroless nickel deposits. As can be seen, the corrosion rates for all coatings (50-65 µm) tested increased with time, the greatest attack occurring on medium and low-phosphorus deposits. High phosphorus electroless nickel showed better results, but would still be expected to fail under a constant immersion situation. Consequently, as in the case for food processing equipment exposed to phosphoric acid, periodic wash down procedures would be necessary to achieve an acceptable service life. Furthermore, since nickel dissolution occurs under these conditions, no passive films remain on the coating surface. Therefore, accelerated attack of the base metal can be anticipated if pores exist in the deposit.

Tests were also conducted on the medium phosphorus and high phosphorus deposits in both hydrofluoric acid (15% by volume) and hydrochloric acid (15%) for 7 days at 38°C (93°F). In each solution, accelerated attack of the base metal was severe but the high-phosphorus content deposit maintained its integrity much longer. This again reinforces the conclusion that high-phosphorus deposits offer improved mineral acid resistance. Referring back to the metallurgical data, the fact that the high phosphorus electroless nickel deposit has a high nickel content surface layer may influence corrosive activity in addition to its porosity.

Interestingly, there are practical commercial applications, where electroless nickel is used in aggressive acid service. These include well completion and stimulation applications where periodic fracturing of well formations with HF or HCl is necessary. These procedures are generally followed by flushing of the system to avoid a constant acid environment in contact with the plated steel.

Corrosion tests have also been conducted in a number of other acidic and alkaline environments found in various process industries to assess the suitability of electroless nickel as a replacement for traditional, more expensive materials. In one series of tests, immersion studies were performed according to ASTM G-31, "Laboratory Immersion Corrosion Testing of Metals." Table 3 summarises chemicals tested in this study together with some of their common industry applications. Test panels of 1010 carbon steel were plated with the three different electroless nickel coatings to a thickness of 50-65 microns (2-2.6 mil). All coupons included in the corrosion tests were examined according to ASTM B733 9.61 ferroxyl test to ensure the coatings were free of porosity.

Results of the 60-day weight loss corrosion tests reported in Table 4 provide a new perspective to the commonly held view that high phosphorus deposits are the most corrosion resistant. As can be seen, a mixed set of results is achieved for the different coatings depending on the particular environment involved. In particular, the low phosphorus deposit offers superior corrosion resistance in orthochlorobenzyl chloride, benzotrichloride, benzoyl chloride, and sodium hydroxide. This suggests that successful application of electroless nickel as a corrosion resistant

**Table 4. Weight Loss Corrosion Rates of Several Electroless Nickel Deposits in Various Chemical Processing Environments**

| *60 Days, 40°C ± 2°C, 50 µm Deposit* | | | |
|---|---|---|---|
| *Corrosion Rates µm/year* | *LP* | *MP* | *HP* |
| Thionyl Chloride | 90.0 | 1.8 | 2.5 |
| Orthochlorobenzyl Chloride (Crude) | 3.8 | 7.4 | 7.1 |
| Phosphoric Acid | 90.0 | 19.3 | 19.3 |
| Benzotrichloride | 2.5 | 5.6 | 6.1 |
| Benzoyl Chloride | 1.0 | 0.8 | 0.5 |
| Phosphorus Oxychloride | 28.4 | 1.5 | 2.5 |
| Sodium Hydroxide | | | |
| 45% NaOH, 5% NaCL, 40°C | 0.3 | 0.3 | 0.8 |
| 45% NaOH, 5% NaCL, 140°C | 5.3 | 11.9 | Failed |
| 35% NaOH, 93°C | 5.3 | 17.8 | 13.2 |
| 50% NaOH, 93°C | 6.1 | 4.8 | 9.4 |
| 73% NaOH, 120°C | 2.3 | 7.4 | Failed |

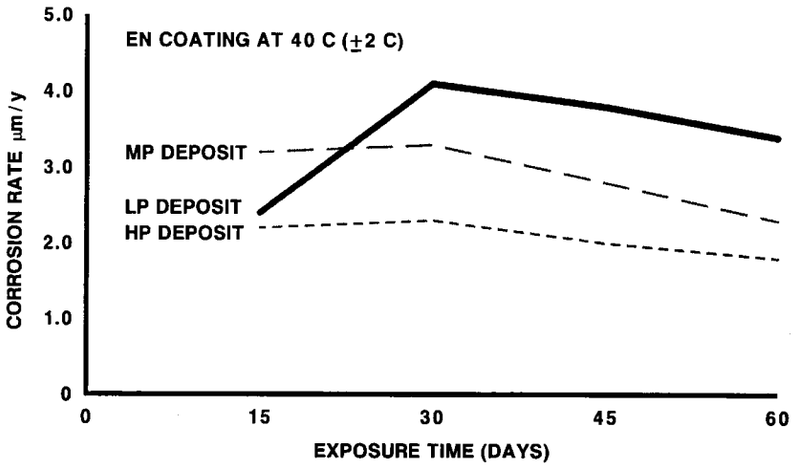
material in a specific industrial process environment requires consideration of more than one deposit type. Examination of corrosion rates over a 60-day period for the electroless nickel deposits tested reveals some interesting trends. Figures 10-13 show that corrosion rates, in nearly all cases, decline or stabilise with time. This results directly from the formulation and maintenance of protective passive films on the deposit surface within the given environment. For phosphorus oxychloride and benzotrichloride, behavioural differences between the LP and HP deposit are apparent. The slightly acidic phosphorus oxychloride, as expected, exhibited low corrosivity toward higher phosphorus content deposits. In contrast, benzotrichloride, a non-oxidizing, neutral pH solvent was less aggressive toward the low phosphorus deposit.

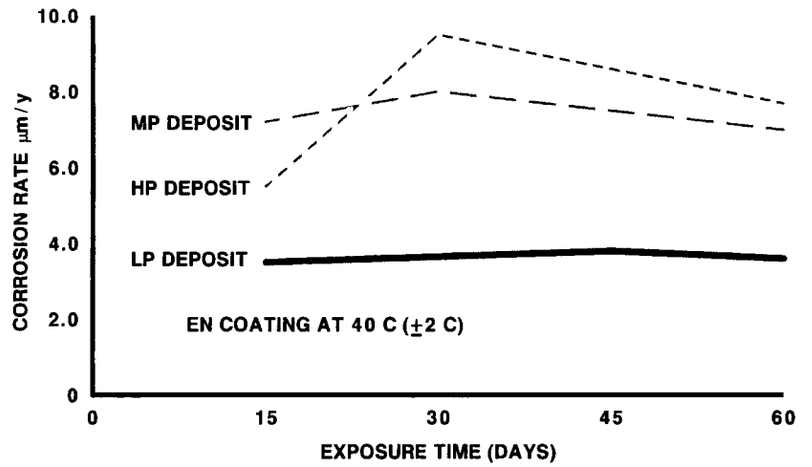
Highly alkaline environments, i.e. sodium hydroxide are prevalent in most process industry production environments. Food processing, textile manufacture, pulp and paper production, petroleum refining are just a few examples. The results of corrosion testing in sodium hydroxide from 35% to 73% by weight are summarised in Table 4. Figures 14-16, however, reveal the extent of the performance differences between different electroless nickel coatings by providing a comparison of corrosion rates over the 60-day test period. The low-phosphorus alloy, exhibited superior elevated temperature performance at 60°C (140°F), 45% sodium hydroxide concentration (Figure 14) and 49°C (120°F), 73% sodium hydroxide (Figure 15.). Furthermore, stable corrosion rates over the 60-day period indicated the protective nature of the oxide film formed on the nickel phosphorus deposit. In contrast, the high phosphorus deposit at elevated temperatures in caustic environments failed almost immediately (Table 4). The medium phosphorus deposit interestingly, followed closer behind the low phosphorus deposit with somewhat higher corrosion rates. In sodium hydroxide 50%, the low-phosphorus deposit was still superior to the others, although in this case, the other coatings showed improved corrosion rates. This suggests that under less aggressive conditions passive films can be sustained on these coatings as well.

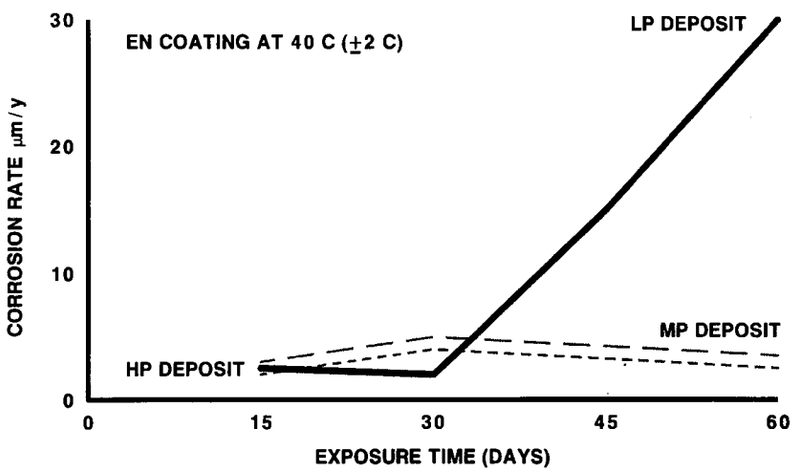
In many of these applications, electroless nickel competes against stainless steel, certain non-ferrous alloys like Nickel 200, or uncoated steel. (The corrosion resistance of electroless nickel, compared to these alternate materials, is summarised in Table 5). By matching the correct electroless nickel deposit to a particular chemical environment, these data suggest that electroless nickel is a viable alternative to more costly high alloy materials such as Nickel 200 or AISI 316 stainless steel.

## LOW PHOSPHORUS PROCESS CHARACTERISTICS

The introduction of the unique chemistry of lower phosphorus electroless nickel products to the marketplace brings with it some very significant improvements in processibility for job shops as summarised in Table 6. For two different low phosphorus



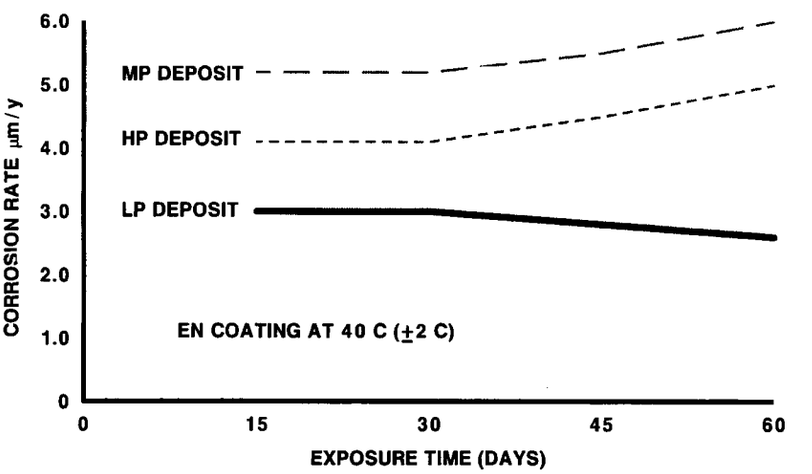




*Figure 10. Corrosion performance of thionyl chloride 60 day weight loss test.*
*Figure 11. Corrosion performance of Ocbo (crude) 60 day weight loss test.*
*Figure 12. Corrosion performance of phosphorus oxychloride 60 day weight loss test.*
*Figure 13. Corrosion performance of BTC 60 day weight loss test.*

**Table 5. Comparison of Weight Loss Corrosion Results of Electroless Nickel to Alternate Uncoated Materials**

| *60-Day, 40°C ± 2°C* | *Nickel 200* | *Mild Steel* | *AISI 316 SS* | *Corrosion Rate EN Coated 1010-50μm* | *Suggested Deposit Type* |
|---|---|---|---|---|---|
| Thionyl Chloride | 7.0 | 200.0 | 5.1 | 1.8-2.5 | MP/HP |
| Orthochlorobenzyl Chloride | 12.0 | NA | 25.0 | 3.8 | LP |
| Phosphoric Acid | 10.0 | 1270.0 | 2.5 | 1.9-3 | None |
| Benzotrichloride | 5.1 | 9.0 | 5.1 | 2.5 | LP |
| Benzoyl Chloride | 5.1 | 8.6 | 5.1 | 0.5-1.0 | LP/MP/HP |
| Phosphorus Oxychloride | 10.1 | 100.0 | 18.8 | 1.5-2.5 | MP/HP |
| Sodium Hydroxide | | | | | |
| 45% NaOH, 5% NaCl, 40°C | 2.5 | 35.6 | 6.4 | 0.3-0.5 | LP/MP/HP |
| 45% NaOH, 5% NaCl, 140°C | 80.0 | NA | 27.9 | 5.3 | LP |
| 35% NaOH, 93°C | 5.1 | 94.0 | 52.0 | 5.3 | LP |
| 50% NaOH, 93°C | 5.1 | 533.4 | 83.8 | 4.8-6.1 | LP/MP |
| 73% NaOH, 120°C | 5.1 | 1448.0 | 332.7 | 2.3 | LP |

**Table 6. A Comparison of Low Phosphorus EN to Conventional EN Plating Processes**

| | *1-3% P LP EN* | *3.5-4.5% P LP EN* | *6-8% P MP EN* | *10-12% P HP EN* |
|---|---|---|---|---|
| pH | | | | |
| Range | 5.8-6.8 | 5-8 | 4.5-5.2 | 4.4-5.0 |
| Optimum | 6.2 | 6.2-6.8 | 4.8 | 4.6-4.8 |
| Temp., °F, | | | | |
| Range | 145-195 | 140-190 | 180-195 | 180-195 |
| Optimum | 180-190 | 175 | 190 | 185-190 |
| Plating Rate | | | | |
| Range | | | | |
| New Bath | 0.6-0.8 (190°F)<br>0.4-0.6 (180°F) | 0.8-1.0 (190°F)<br>0.5-0.7 (175°F) | 0.8-0.9 | 0.5-0.6 |
| Aged Bath | | | | |
| 6-8 mto | 0.5-0.6 | 0.4-0.8 | 0.6-0.8 | 0.4-0.5 |
| Appearance | Semi-bright | V-bright | V-bright | Semi-bright |
| Recommended Agitation | Solution | Solution | Mild Air | Mild Air or Solution |
| Recommended Work Load $dm^2/l$ | > 0.5 $dm^2/l$ | < 0.2 $dm^2/l$ or lower | > 0.2 $dm^2/l$ | < 0.2 $dm^2/l$ |

deposits, there are some significant variations in plating characteristics when compared to conventional electroless nickel coatings.

One of the most significant characteristics of the two low phosphorus processes is the wide operating range afforded by these products and the capability of this family of products to plate over a very wide range of temperature and pH, i.e., 60-90°C (140-195°F). Thus, the flexibility of LP processes has important implications for the plater. First, these products allow plating to begin much earlier during heat up of the plating bath. At the beginning of the week, after the electroless tank has been shut down over the weekend, for example, problems with heat up of the plating bath can delay production or require someone to come in hours ahead of time to get the bath up to temperature. Using one of these processes you can begin plating even at 60-65°C (140°-150°F) as long as the bath is properly replenished and an adequate among of work is in the tank. Platers with very large tanks can benefit from this since heat up time of 4-6 hours to reach 82-88°C (180°-190°F) are not uncommon. Low phosphorus processes can cut this time by 50% or more. The ability of this process to avoid step plating when the bath is inadvertently operated at lower temperatures is also a plus. One of the most significant benefits of this new electroless chemistry is the potential energy savings associated with being able to plate at lower temperatures. Again, this is of particular importance to platers who have larger tanks and find it difficult and expensive to maintain operating temperatures of 85-90°C (185-195°F) typically required for conventional electroless nickel baths.

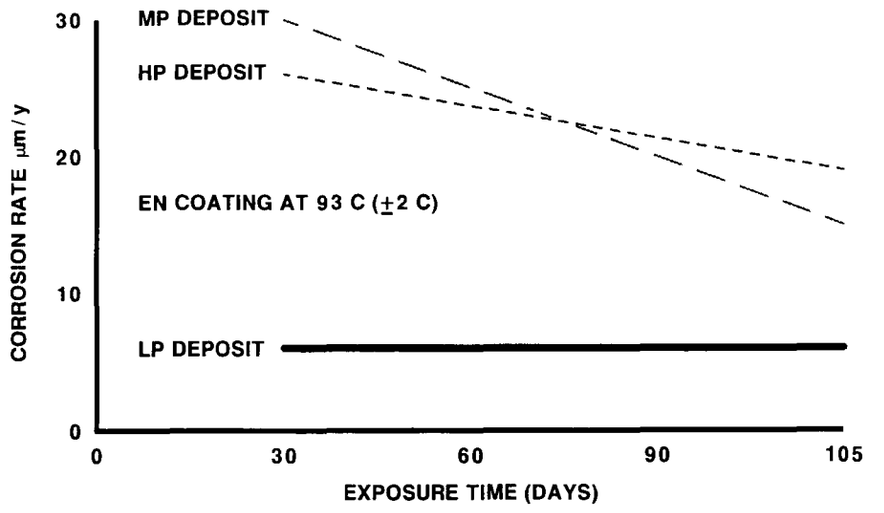
The variability of pH is also a benefit in certain situations for low phosphorus electroless nickel process. Substrates such as certain plastics and ceramics which are sensitive to pH (and temperatures) can be readily accommodated with the same bath that is used for plating metals. Furthermore, plating rate and deposit compositions can be varied more widely by adjusting either pH or temperature. During bath operation, the bath color will change from blue to green indicating a decrease in the pH value. pH adjustment must be made to return the bath to an optimum pH level.

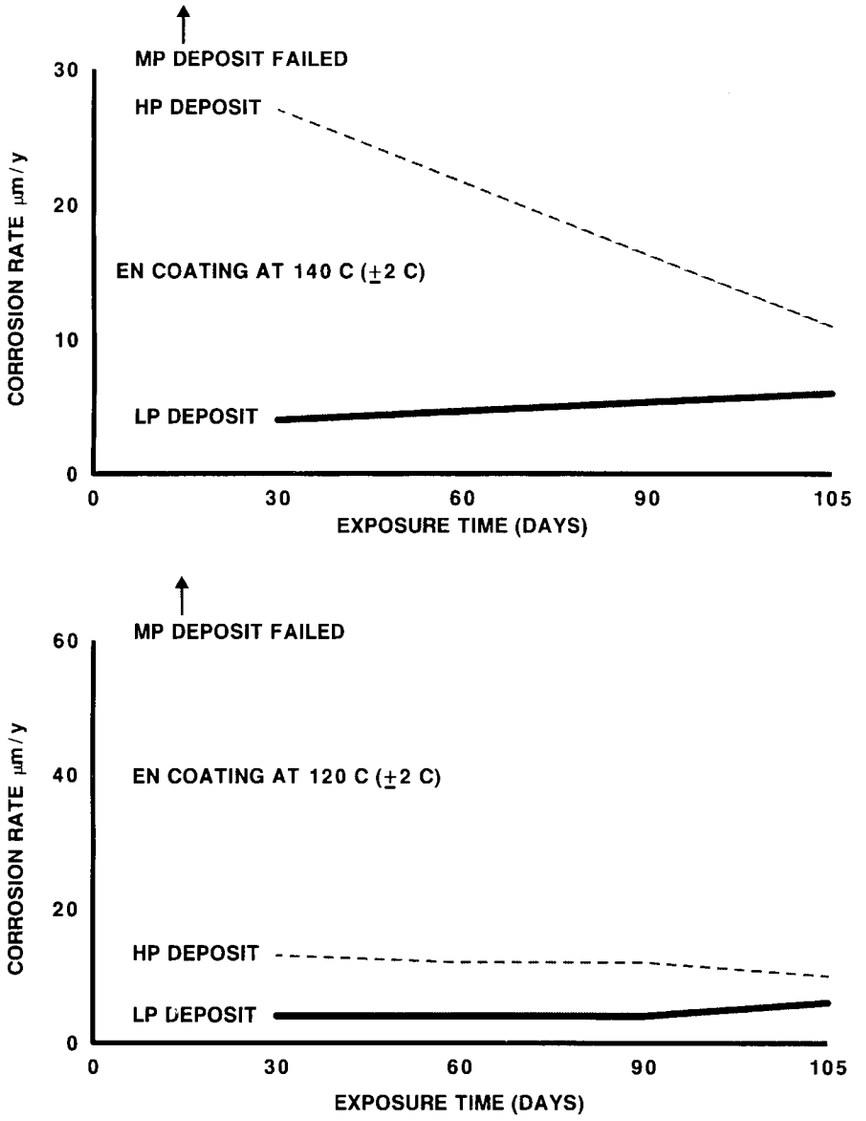
Although the technical data presented in this paper is based on a 1-3% phosphorus deposit, the technology is capable of retaining many of the characteristic engineering properties while offering alternative processing and deposit characteristics to meet alternative application requirements. Comparing low phosphorus process 1 and 2 in Table 6, for instance, the second process has advantages in deposit brightness, low work load plating capacity, high deposition rate, its ability to plate at a fast rate at only 80°C (175°F), and being formulated for less porosity and better corrosion protection. Some of the engineering deposit properties for this process are not the same as those for the 1-3% P deposit, however.

## CONCLUSION

Low phosphorus electroless nickel plating processes offer advantages in their engineering properties and operating flexibility that deserve special attention by the plating industry. Furthermore, this technology opens up new market applications for electroless plating not available with present products on the market.

*Figure 14. Sodium hydroxide (35%) corrosion rate*






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*Figure 15. Above: corrosion performance of sodium hydroxide (45%) 60 day weight loss test.*
*Figure 16. Below: corrosion performance of sodium hydroxide (73%) 60 day weight loss test.*